

A Proof of Hasse-Minkowski

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This writeup will work towards a proof of the following theorem, and some of its corollaries.

Theorem 1 (Hasse-Minkowski). If f is a quadratic form, then it is equivalent for f to represent 0 over $\mathbb{Q}$ and over all of $\mathbb{Q}$'s completions.

Definition 2. Let V be a module over a commutative ring A . We say $Q : V \rightarrow A$ is a quadratic form if

$$Q(ax) = a^2Q(x) \quad \text{and} \quad (x, y) \mapsto Q(x + y) - Q(x) - Q(y) \text{ is bilinear.}$$

Definition 3. We say that Q represents a if there is a non-zero element x with $Q(x) = a$.

1 Introduction

1.1 Initial definitions

Observation 4. If we restrict ourselves to the case of A is a field, call it k , with non-2 characteristic, and V is finite dimensional, then we can define

$$x \cdot y = \frac{1}{2}(Q(x+y) - Q(x) - Q(y)).$$

Observe that $Q(x) = x \cdot x$, so this gives us a bijection between quadratic forms of symmetric bilinear forms.

Definition 5. We say that x, y are *orthogonal* if $x \cdot y = 0$. We write $H^\perp$ to be everything orthogonal to everything in H .

Fix some basis e_i of V , then if we define $A = (e_i \cdot e_j)_{ij}$ then a change of basis scales A by some square. Thus, we have a natural map from the set of quadratic forms to $k/(k^\times)^2$ via determinant up to change of basis.

Proposition 6. Say Q is non-degenerate if $V^\perp$ is trivial. We have Q non-degenerate iff its discriminant is non-zero.

Proposition 7. Suppose (V, Q) is non-degenerate. Then $\dim U + \dim U^\perp = \dim V$.

Proof. We have an exact sequence

$$0 \rightarrow U^\perp \xrightarrow{\psi} V \xrightarrow{\varphi} U^* \rightarrow 0$$

with the ψ being inclusion and φ assigning being the function that takes $v \in V$ to $y \mapsto v \cdot y$. This is surjective because it is also the composition of the natural surjections $V \rightarrow V^*$ and $V^* \rightarrow U^*$. By the First Isomorphism Theorem (rank-nullity), we have $\dim U^* + \dim U^\perp = \dim U + \dim U^\perp = \dim V$. $\square$

Observe that if U is one-dimensional then $U^\perp$ is a hyperplane.

Definition 8. $x \in V$ is called *isotropic* if $Q(x) = 0$. We say a subspace U is *isotropic* if all of its elements are isotropic.

Note. f represents 0 iff the corresponding module contains a non-zero isotropic element.

Proposition 9. If (V, Q) is non-degenerate and contains a non-zero isotropic element x then $Q(V) = k$.

Proof. Consider $z \in V$ with $x \cdot z = 1$. Then $y = z - \frac{z \cdot z}{2}x$ is isotropic because

$$Q(y) = -\frac{z \cdot z}{2}2(z \cdot x) + Q(z) + Q(-\frac{z \cdot z}{2}x) = 0.$$

If $a \in k$ then $\frac{a}{2} = x \cdot (\frac{a}{2}z) = \frac{1}{2}\left(Q(x + \frac{a}{2}z) - Q(x) - Q(\frac{a}{2}z)\right)$, so $a = Q(x + \frac{a}{2}z)$. $\square$

Theorem 10. Every quadratic module has an orthogonal basis.

Proof. We will induct on $n = \dim V$. Observe that if $n = 0$ of V is isotropic then we are done. Otherwise, let $e_1 \cdot e_1 \neq 0$. The orthogonal complement of e_1 has dimension $\dim V - 1$ and thus by the induction hypothesis has orthogonal basis $(e_2, e_3, \dots, e_n)$ and thus $(e_1, e_2, \dots, e_n)$ is a orthogonal basis of V (we have $e_i \cdot e_1 \neq 0$ with $i > 1$ because they are orthogonal). $\square$

Theorem 11 (Witt). Let (V, Q) and (V', Q') be nondegenerate and $U \subsetneq V$. Let $s : U \rightarrow V'$ be injective, show that it can be extended to an isomorphism from V to V' .

The proof is omitted; it is similar to the same result over a normal vector space.

1.2 Translations

Definition 12. Let A be a $n \times n$ symmetric matrix. We say (k^n, f) is a quadratic module associated to f or A where

$$f(X) = \sum_{i=1}^n \sum_{j=1}^n a_{ij} X_i X_j.$$

Notation 13. We write $f \sim f'$ if there is some invertible X with $A = XA'X^T$.

Notation 14. Let $f(X_1, \dots, X_n)$ and $g(X_{n+1}, \dots, X_{n+m})$ be two forms, then we write $f + g$ for the quadratic form $f(X_1, \dots, X_n) + g(X_{n+1}, \dots, X_{n+m})$ and we write $f - g$ for $f + (-g)$.

Proposition 15. TFAE for g non-degenerate $a \in k^\times$.

1. g represents a
2. $g \sim h + aZ^2$
3. $f = g - aZ^2$ represents 0

Proof. 2 implying 1 and 3 is obvious. Now suppose 1. We have some x with $x \cdot x = a$, and let H be the orthogonal complement of x and h its associated quadratic form. Then $V = H \oplus kx$ and thus $g \sim h + aZ^2$. Suppose 3, we have $(x_1, x_2, \dots, x_n, z)$ representing 0. If z is non-zero, then $g(x_1, x_2, \dots, x_n) = az^2$ and thus $g(x_1/z, x_2/z, \dots, x_n/z) = a$. If $z = 0$ then g represents 0 by Proposition 9 we have that $g(V) = k \ni a$, so we have 1. Hence $2 \implies 3 \implies 1 \implies 2$. $\square$

Definition 16. The rank of Q is the codimension of $V^\perp$ and is equal to $\dim V - \dim V^\perp$.

Proposition 17. Let g, h be non-degenerate with positive rank and let $f = g - h$. TFAE

1. f represents 0
2. There is some $a \in k^\times$ which is represented by g and h
3. There is some $a \in k^\times$ such that $g - aZ^2$ and $h - aZ^2$ represent 0

Proof. $3 \iff 2 \implies 1$ is trivial. Now suppose $g(x) - h(y) = 0$ with $(x, y) \neq 0$. WLOG assume that $x \neq 0$. Then either $g(x) = 0$ and thus by Proposition 9 g is surjective and because h takes on a non-zero value we are done or $g(x) \neq 0$ and thus 2 is trivially satisfied. $\square$

Corollary 18. By Theorem 10 there exists some $a_1, \dots, a_n$ with $f \sim a_1 X_1^2 + \dots + a_n X_n^2$.

The rank is thus the number of i with a_i nonzero, and f is full rank iff f is non-degenerate.

Corollary 19. By Theorem 11 if $f = g + h$ and $f' = g' + h'$ with $f \sim f'$ and $g \sim g'$ then $h \sim h'$.

2 Hilbert Symbol

Definition 20 (Hilbert symbol). Over some field k and for $a, b \in k$ we set $(a, b) = 1$ if

$$\exists(x, y, z) \in k^3 \text{ such that } z^2 = ax^2 + by^2 \text{ and } (a, b, c) \neq (0, 0, 0).$$

Proposition 21. $(a, b) = 1$ iff a is a norm of an element of $k(\sqrt{b})$.

Proposition 22. For all $t \in k^\times$ we have $(t, t) = (-1, t)$.

Proof. Suppose $z^2 = tx^2 + ty^2$. Then $tz^2 - (tx)^2 = (ty)^2$, so $(t, -1) = 1$. Now suppose $z^2 = -x^2 + ty^2$. Then $tz^2 + tx^2 = (ty)^2$ so $(t, t) = 1$. Hence $(t, -1) = 1 \iff (t, t) = 1$, so $(t, -1) = (t, t)$. $\square$

Corollary 23. The Hilbert symbol is symmetric and linear in both coordinates as a map from $k^\times \times k^\times \rightarrow \pm 1$.

Let $\mathbf{e} = (e_1, e_2, \dots, e_n)$ be an orthogonal basis of V . Let $a_i = e_i \cdot e_i$ and then we have

$$d(Q) = \prod_{i=1}^n a_i \in k^\times / (k^\times)^2.$$

Definition 24.

$$\varepsilon(\mathbf{e}) = \prod_{i < j} (a_i, a_j) = \prod_{i \leq j} (a_i, a_j).$$

Theorem 25. ε is independent of basis choice.

Proof. Fix an n and a basis e .

If $n = 1$ then $\varepsilon(e) = 1$. Similar, if $n = 2$, we have $\varepsilon(e) = 1$ iff $a_1X^2 + a_2Y^2$ represents 1, iff there is some v with $Q(v) = 1$, which is independent of e .

Now, we proceed by induction. Pick a different orthonormal basis e' . It suffices to show the result when e' and e share an element in common, or $e'_1 = e_1$. If $a_i = e_i \cdot e_i$ and likewise $a'_i = e'_i \cdot e'_i$, we then have

$$\varepsilon(e) = (a_1, \prod_2^n a_i) \prod_{2 \leq i < j} (a_i, a_j) = (a_1, a_1 d(Q)) \prod_{2 \leq i < j} (a'_i, a'_j) = (a_1, \prod_2^n a'_i) \prod_{2 \leq i < j} (a'_i, a'_j) = \varepsilon(e')$$

because $d(Q)$ is an invariant and we use the inductive hypothesis on the orthogonal component to $e_1 = e'_1$. $\square$

Notation 26. We can now write $\varepsilon(f)$ or $\varepsilon(Q)$ because it is well-defined.

3 Quadratic forms over $\mathbb{Q}_p$

We will fix a prime p and assume nondegeneracy for this section.

Theorem 27. Let p be an odd prime. For $x = p^n u \in \mathbb{Q}_p^\times$ we have x a square iff n is even and $\bar{u} \in \mathbb{F}_p^\times$ is a square.

Proof. For the backwards direction, by Hensel lifting, since k is a root of $x^2 - u$ for some $k \in \mathbb{F}_p$, we will have a square root of u in $\mathbb{Q}_p^\times$. For the forwards direction, if $x^2 - u$ has a root in $\mathbb{Q}_p$ then it has a root up to $p\mathbb{Z}_p$, and when we quotient by $p\mathbb{Z}_p$ we get $\mathbb{F}_p$, and thus $\bar{u}$ is a square in $\mathbb{F}_p^\times$. $\square$

Theorem 28. Let $p = 2$. For $x = p^n u \in \mathbb{Q}_p^\times$ we have x a square iff n is even and $u \equiv 1 \pmod{8}$.

Proof. Let $a \equiv 5 \pmod{8}$. Define $\theta_{n,\alpha} : \mathbb{Z}/(2^{n-2}\mathbb{Z}) \rightarrow (1 + 4\mathbb{Z}_2)/(1 + 2^n\mathbb{Z}_2)$ via $z \mapsto (\alpha_n)^z$. If we take a direct limit of both sides, we get an isomorphism $\mathbb{Z}_2 \rightarrow (1 + 4\mathbb{Z}_2)$. Hence, squares in $\mathbb{Z}_2$ are elements in $1 + 8\mathbb{Z}_2$, so $u \equiv 1 \pmod{8}$.

For the backwards direction, observe that in $\mathbb{Z}/2^n\mathbb{Z}$ with $n \geq 3$ needs to have 2^{n-3} squares, and squares must be 1 mod 8, and thus everything 1 mod 8 is a square. Hence, anything 1 mod 8 is a square in the limit. Thus, we are done. $\square$

Lemma 29. If $a \neq 1 \in (k^\times)/(k^\times)^2$ then there are an equal number of elements $x \in (k^\times)/(k^\times)^2$ with $(x, a) = 1$ and $(x, a) = -1$. Further, if solutions to $(x, a) = \varepsilon$ and $(x, a') = \varepsilon'$ are disjoint then $a = a'$ and $\varepsilon = -\varepsilon$.

The proof of this is omitted.

Theorem 30. f represents 0 in $\mathbb{Q}_p$ iff

1. $n = 2$ and $d = -1 \in \mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^2$
2. $n = 3$ and $(-1, -d) = \varepsilon$
3. $n = 4$ and either $d = 1$ or $\varepsilon = -(-1, -1)$
4. $n \geq 5$

Proof. We will deal with these cases one by one. It turns out that case (4) is only necessary for Corollary 38, and hence is proved in the Appendix.

(1):

f represents 0 iff $-a_1/a_2$ is a square. In $\mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^2$ we have $-a_1/a_2 = -a_1a_2 = -d$, so $d = -1$.

(2):

We have $-a_3f \sim -a_3a_1X_1^2 - a_3a_2X_2^2 - a_3a_3X_3^2 \sim -a_3a_1X_1^2 - a_3a_2X_2^2 - X_3^2$. From the definition of the Hilbert symbol, the right side represents 0 iff

$$\begin{aligned} 1 &= (-a_3a_1, -a_3a_2) = (-1, -1)(-1, a_3)(-1, a_2)(a_3, -1)(a_3, a_3)(a_3, a_2)(a_1, -1)(a_1, a_3)(a_1, a_2) \\ &= (-1, -1)(-1, a_2)(a_3, a_3)(a_1, -1)(a_1, a_3)(a_1, a_2)(a_3, a_2) \\ &= (-1, -a_1a_2)(a_3, a_3)\varepsilon. \end{aligned}$$

Hence, by Proposition 22 we are done.

(3):

By Proposition 17, f represents 0 iff there is some $x \in \mathbb{Q}_p^\times$ which is represented by both $a_1X_1^2 + a_2X_2^2$ and $-a_3X_3^2 - a_4X_4^2$. By Lemma 29 this is impossible exactly when $a_1a_2 = (-a_3)(-a_4) = a_3a_4$ and $(a_1, a_2) = -(-a_3, -a_4)$. Hence, $d = a_1a_2/(a_3a_4) \cdot (a_3a_4)^2 = 1 \in (\mathbb{Q}_p^\times)/(\mathbb{Q}_p^\times)^2$. We have

$$\varepsilon = (a_1, a_2)(a_1, a_3)(a_1, a_4)(a_2, a_3)(a_2, a_4)(a_3, a_4) = (a_1, a_2)(a_3, a_4)(a_1, a_3a_4)(a_2, a_3a_4).$$

further, we have

$$\varepsilon = (a_1, a_2)(a_3, a_4)(a_1a_2, a_3a_4) = (a_1, a_2)(a_3, a_4)(a_3a_4, a_3a_4) = -(-a_3, -a_4)(a_3, a_4)(-1, a_3a_4)$$

and thus

$$\varepsilon = -(-a_3, -a_4)(a_3, a_4)(-1, a_4)(-1, -a_3) = -(-a_3, -a_4)(-a_3, a_4)(-1, a_3) = -(-1, a_3)(-a_3, -a_4^2).$$

Finally, we have

$$\varepsilon = -(-1, a_3)(-a_3, -1) = -(-1, -a_3^2) = -(-1, -1). \quad \square$$

Corollary 31. f represents $a \in \mathbb{Q}_p^\times$ exactly when

1. $n = 1$ and $a = a_1 = d$
2. $n = 2$ and $(a, -d) = \varepsilon$.
3. $n = 3$ and either $a = -d$ or $(-1, -d) = \varepsilon$.
4. $n \geq 4$

4 Combining our results

We will let V be the set of primes and infinity. Fix a non-degenerate quadratic form $f \sim \sum_{i=1}^n a_i X_i^2$.

Definition 32. The discriminant $d(f) \in \mathbb{Q}^\times / (\mathbb{Q}^\times)^2$ is $\prod_i a_i$.

Observation 33. If ι_v is the natural injection $\mathbb{Q} \rightarrow \mathbb{Q}_v$, then $d_v(f) = \iota_v(d(f))$.

Lemma 34. For any f we have

$$\prod_{v \in V} \varepsilon_v(f) = 1.$$

Proof. Observe that it suffices to show that

$$\prod_{v \in V} (a, b)_v = 1.$$

This proof is in the Appendix as Theorem 61. $\square$

Theorem 35. Let $(a_i)_{i \in I}$ be a finite set of elements of $\mathbb{Q}^\times$ and $(\varepsilon_{i,v})_{i \in I, v \in V}$ be ± 1 . For there to be some x with $(a_i, x)_v = \varepsilon_{i,v}$ the following conditions are necessary and sufficient.

1. A cofinite number of $\varepsilon_{i,v}$ are 1
2. We have $\prod_V \varepsilon_{i,v} = 1$ for all i
3. There is some x_v with $(a_i, x_v)_v = \varepsilon_{i,v}$

Observe that condition 1 and 2 are given by Theorem 61, and condition 3 is trivially necessary.

Proof. WLOG take a_i to be square free integers. Let $S \subset V$ of $\infty, 2$ and prime factors of any a_i and T be the set of $v \in V$ with $\varepsilon_{i,v} = -1$. Observe that by condition 1 we have S and T finite.

Suppose $S \cap T = \emptyset$. Let

$$a = \prod_{\ell \in T, \ell < \infty} \ell \text{ and } m = 8 \prod_{\ell \in S, 2 < \ell < \infty} \ell.$$

By Corollary 57 there is some $p \equiv a \pmod{m}$ not in either S or T . Then ap satisfies all the conditions we want (verification is omitted).

If $S \cap T$ is not empty. Take some $x' \in \mathbb{Q}$ with x'/x_v a square for $v \in S$. If we now replace all of the x_v s with x' for $v \in S$ then we have S and T disjoint, so we fall into case 1. $\square$

4.1 Proof

Theorem 36 (Hasse-Minkowski). If f is a quadratic form, then it is equivalent for f to represent 0 over $\mathbb{Q}$ and over all of $\mathbb{Q}$'s completions.

The necessity is trivial. We split the proof of sufficiency up based on n .

Proof when $n \leq 2$. When $n \leq 1$, f does not represent 0 over $\mathbb{Q}$ or $\mathbb{R}$. When $n = 2$, we have $f = X_1^2 - aX_2^2$ with $a \in \mathbb{Q}$. Since f_∞ represents 0 we have $a > 0$. To show that a is a square, it suffices to show that a has even valuation mod every p . However, in $\mathbb{Q}_p$, we know that a is a square, and thus by Theorem 27 and Theorem 28 we are done. $\square$

Proof when $n = 3$ (Legendre). Let $f = X^2 - aY^2 - bZ^2$. WLOG we can assume a and b are square free integers with $|a|_\infty \leq |b|_\infty$. We will proceed by induction on $m = |a|_\infty + |b|_\infty$. The minimal value that m can take on is 2, and if $a = b = -1$ then f does not represent 0 over $\mathbb{Q}$ or $\mathbb{R}$ and if either is positive then we clearly represent 0 over $\mathbb{Q}$.

We now proceed by induction. Assume $m > 2$, and thus $|b|_\infty > 2$. Write $b = \pm p_1 p_2 \cdots p_k$. We will first show that if $p = p_i$ then a is a square mod p .

If $p \mid a$ this is obvious. Because f represents 0 in $\mathbb{Q}_p$ we can find x, y, z all with non-negative valuation and not all positive valuation (by multiplying by an appropriate power of p). Hence, mod p we have $x^2 - ay^2 \equiv bz^2 \equiv 0$ and thus a is a square or $x \equiv 0 \implies y \equiv 0 \implies z \equiv 0$, but they x, y, z all have positive valuation. Hence a is a square mod p , and by the Chinese Remainder Theorem is a square mod b .

Hence, there is some $t, b' \in \mathbb{Z}$ with $t^2 = a + bb'$. We can, we can pick t such that $|t|_\infty \leq |b|_\infty/2$. Thus $(bb', a)_v = 1$ and thus $(b, a)_v = (b', a)_v$. Hence, $f' = X - aY^2 - b'Z^2$ has represents 0 iff f does. We have

$$|b'| = \left| \frac{t^2 - a}{b} \right| \leq \frac{|b|}{4} + \frac{|a|}{|b|} \leq \frac{|b|}{4} + 1 < |b|$$

with the last inequality because $|b| > 2$. Hence, if we find $c \mid b'$ square-free with $c\tau^2 = b'$, then $|c| < |b|$, so by induction f' represents 0 in $\mathbb{Q}$, and thus f does. $\square$

Proof when $n = 4$. Let $f = aX_1^2 + bX_2^2 - (cX_3^2 + dX_4^2)$. By Proposition 17, for each v we have some $x_v \in \mathbb{Q}_v^\times$ which is represented by $g = aX_1^2 + bX_2^2$ and $h = cX_3^2 + dX_4^2$. By Corollary 31, we have

$$(x_v, -ab)_v = (a, b)_v, (x_v, -cd)_v = (c, d)_v.$$

By Theorem 35, we have some $x \in \mathbb{Q}^\times$ with

$$(x, -ab)_v = (a, b)_v, (x, -cd)_v = (c, d)$$

for all v . Hence, $aX_1^2 + bX_2^2 - xY^2$ represents 0 in each completion of $\mathbb{Q}$, and thus in $\mathbb{Q}$ by the $n = 3$ proof. Hence, g represents x . By the same argument we have that h represents x and thus we are done. $\square$

The proof for $n \geq 5$ mimics the proof for $n = 4$, with a little induction added in.

Proof when $n \geq 5$. Let $f = g - h$ with

$$g = a_1X_1^2 + a_2X_2^2, h = -(a_3X_3^2 + \cdots + a_nX_n^2).$$

Let S be $2, \infty$ and all the primes dividing some a_i with $3 \leq i \leq n$. There is some a_v that is represented by g and h . Like with the previous case, take $x_1, x_2 \in \mathbb{Q}$ with $a = g(x_1, x_2)$ and a/a_v squares in $\mathbb{Q}_v$ for all $v \in S$. Consider $\tilde{f} = aZ^2 - h$. This represents $0 \in \mathbb{Q}_v$. When $v \notin S$, we have each of the a_i a $\mathbb{Z}_p$ unit, and thus $(-1, -d)_v = 1 = \varepsilon_v(h)$, with the last equality holding because each term in the product is 1. Thus by Corollary 31 represents 0. Further, $\tilde{f}$ represents 0 in each $\mathbb{Q}_v$ and thus, by induction, represents 0 in $\mathbb{Q}$. Hence h represents a in $\mathbb{Q}$, and thus f represents 0, and thus we are done. $\square$

5 Other results

Corollary 37. For $a \in \mathbb{Q}$ it is equivalent to f to represent a in $\mathbb{Q}$ and in all of its completions.

Corollary 38. A form of rank ≥ 5 represents 0 iff it represents 0 in $\mathbb{R}$.

Theorem 39 (Gauss). An integer can be written as a sum of three squares iff it is not of the form $4^x(8n-1)$.

Proof. The forwards direction is immediate. For the backwards direction, observe that $a = 4^x(8n-1)$ iff $-a$ is a square in $\mathbb{Q}_2$.

Now consider $a \in \mathbb{Q}^\times$. We know that a is represented by $f = X_1^2 + X_2^2 + X_3^2$ iff $a > 0$ and $(-1, -d)_v = \varepsilon$. We have $(-1, -1) = 1 = \varepsilon$ for odd primes p , but -1 for $p = 2$. Hence $a = -1$ in $\mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^2$, which is to say that $-a$ is a square. This shows that every a is represented by rationals, but does not show that it is represented by integers, which is what we want.

Observe that for $x \in \mathbb{Q}^3$ there is some $y \in \mathbb{Z}^3$ with $f(x - y) < 1$, by rounding each coordinate. Now suppose $n \in \mathbb{Z}$ is represented by f in $\mathbb{Q}$. Minimize t such that $t^2n = f(x)$ for $x \in \mathbb{Z}^3$. Pick a $y \in \mathbb{Z}^3$ with $x/t = y + z$ with $f(z) < 1$. If $f(z) = 0$ then $z = 0$ and hence $x/t \in \mathbb{Z}^3$, and by minimality we have $t = 1$. Otherwise, let

$$a = (y \cdot y) - n, b = 2(nt - x \cdot y), t' = at + b, x' = ax + by.$$

Then $tt' = t^2f(z)$ and thus $t' = tf(z)$, and hence $t' < t$. However, we have $a, b \in \mathbb{Z}$, and thus $t' \in \mathbb{Z}$, but this contradicts the minimality of t . Hence $t = 1$. $\square$

Corollary 40 (Lagrange). Every positive integer can be written as a sum of four squares.

Proof. Let $n = 4^x m$. If m is a sum of three squares, then multiplying each by 2^x gives n as a sum of three squares. Otherwise, $m \equiv 7 \pmod{8}$ and thus $m - 1$ is a sum of three squares. Thus m is a sum of four squares, and thus multiplying by 2^x gives n as a sum of four squares. $\square$

Corollary 41 (Gauss). Every positive integer is a sum of three triangular numbers.

Proof. Consider x', y', z' with $(x')^2 + (y')^2 + (z')^2 = 8n + 3$. Since squares are 0 or 1 mod 4, we have that $x' = 2x + 1$, $y' = 2y + 1$ and $z' = 2z + 1$. Then

$$\begin{aligned} \frac{x(x+1)}{2} + \frac{y(y+1)}{2} + \frac{z(z+1)}{2} &= \frac{(x + \frac{1}{2})^2 - \frac{1}{4}}{2} + \frac{(y + \frac{1}{2})^2 - \frac{1}{4}}{2} + \frac{(z + \frac{1}{2})^2 - \frac{1}{4}}{2} \\ &= \frac{-3}{8} + \frac{(x')^2}{8} + \frac{(y')^2}{8} + \frac{(z')^2}{8} = \frac{-3}{8} + \frac{1}{8}(8n + 3) = n. \end{aligned} \quad \square$$

6 Dirichlet's Theorem on Arithmetic Progressions

6.1 L function basics

Fix an m and let χ be a character mod m .

Definition 42. If $\gcd(n, m) > 1$, by abuse of notation, define $\chi(n)$ to be 0, otherwise let $\chi(n) = \chi(\bar{n})$. Then

$$L(s, \chi) = \sum_{n=1}^{\infty} \chi(n) n^{-s}.$$

Proposition 43. If $\chi = 1$ then

$$L(s, 1) = F(s)\zeta(s) \text{ such that } F(s) = \prod_{p|m} (1 - p^{-s})$$

and thus $L(s, 1)$ is analytic for $\operatorname{Re}(s) > 0$ except a simple pole at $s = 1$.

Proposition 44. When $\chi \neq 1$ we have

$$L(s, \chi) = \prod_p (1 - \chi(p)p^{-s})^{-1}$$

and this converges absolutely for $\operatorname{Re}(s) > 1$ and converges for $\operatorname{Re}(s) > 0$.

This is clear because partial sums of χ are bounded (by $\varphi(m) \leq m$).

Proposition 45. Let $p \nmid m$. If $f(p)$ is the order of $\bar{p}$ mod m , we have

$$\prod_{\chi} (1 - \chi(p)T) = (1 - T^{f(p)})^{\varphi(m)/f(p)}.$$

Proposition 46. Define

$$\zeta_m(s) = \prod_{\chi} L(s, \chi).$$

We have

$$\zeta_m(s) = \prod_{p \nmid m} (1 - p^{-f(p)s})^{-\varphi(m)/f(p)}.$$

Theorem 47. $L(1, \chi) \neq 0$ for all non-trivial χ .

Proof. We proceed by contradiction. If $L(1, \chi) = 0$ we then have ζ_m holomorphic at $s = 1$, and hence ζ_m would converge for $\operatorname{Re}(s)$ positive by Propositions 44 and 45. Now consider some $p \nmid m$ and $s \in \mathbb{R}$. We have

$$(1 - p^{-f(p)s})^{-\varphi(m)/f(p)} \geq 1 + p^{-\varphi(m)s} + p^{-2\varphi(m)s} + \dots = (1 - p^{-\varphi(m)s})^{-1}.$$

At $s = \frac{1}{\varphi(m)}$ this would tell us that the harmonic series converges. Hence we have that $L(1, \chi) \neq 0$. $\square$

Proposition 48. As $s \rightarrow 1^+$ we have $\sum_p p^{-s} \sim -\log(s - 1)$.

Proof. We have

$$\log(\zeta(s)) = \sum_p -\log(1 - p^{-s}) = \sum_p \sum_{k \geq 1} \frac{1}{k} p^{-ks}$$

by the Taylor series of $\log$. Observe that the $k \geq 2$ sum is bounded. Hence, it suffices to show that the residue of ζ at 1 is 1.

We have

$$\zeta(s) = \frac{1}{s-1} + \sum_{n=1}^{\infty} \left(n^{-s} - \int_n^{n+1} t^{-s} dt \right) = \frac{1}{s-1} + \sum_{n=1}^{\infty} \int_n^{n+1} n^{-s} - t^{-s} dt.$$

Taking magnitudes, we get

$$\left| \zeta(s) - \frac{1}{s-1} \right| \leq \sum_{n=1}^{\infty} |s t^{-s-1}| = \sum_{n=1}^{\infty} |s| n^{-\operatorname{Re}(s)-1}.$$

This sum converges absolutely for $\operatorname{Re}(s) > 0$, and thus we are done. $\square$

Definition 49. Let A be a set of primes. Define the *analytic density* of A to be

$$\lim_{s \rightarrow 1^+} \frac{\sum_{p \in A} p^{-s}}{-\log(s-1)}.$$

Note. The prior two results are true more generally over rings of integers of number fields, where our sum is over prime ideals and p is replaced by the norm of said ideal.

6.2 Initial Lemmas

Definition 50. Let χ be a character mod m . Define

$$f_{\chi}(s) = \sum_{p \nmid m} \chi(p) p^{-s}.$$

This clearly converges for $\operatorname{Re}(s) > 1$

Lemma 51. If $\chi = 1$ then $f_{\chi}(s) \sim -\log(s-1)$ as $s \rightarrow 1^+$ because f_{χ} and $\sum p^{-s}$ differ by a bounded amount.

Lemma 52. If $\chi \neq 1$ then f_{χ} is bounded as $s \rightarrow 1^+$.

Proof. We have

$$\log(L(\chi, s)) = \sum_p -\log(1 - \chi(p)p^{-s}) = \sum_p \sum_{n=1}^{\infty} n^{-1} \chi(p)^n p^{-ns},$$

with the last equality by the Taylor series of $\log$. As $s \rightarrow 1$, because $L(\chi, s) \neq 0$ (Theorem 47) we have the right side bounded. We have

$$\sum_p \sum_{n=1}^{\infty} n^{-1} \chi(p)^n p^{-ns} = \sum_p \sum_{n=2}^{\infty} n^{-1} \chi(p)^n p^{-ns} + \sum_p \chi(p) p^{-s}.$$

Observe that the $n \geq 2$ sum converges absolutely, so this manipulation is valid. Additionally, the $n = 1$ sum (the second sum), which is f_{χ} , is bounded. $\square$

Definition 53. Let $\mathbb{P}_a$ be the set of primes congruent to $a \bmod m$.

6.3 Proof

Theorem 54 (Dirichlet). The analytic density of $\mathbb{P}_a$ is $1/\varphi(m)$.

Proof. Fix an a coprime to m . We will consider the sum

$$g_a(s) = \sum_{\chi} \chi(a)^{-1} f_{\chi}(s)$$

where the sum is over all characters mod m . By the definition of f we have

$$= \sum_{\chi} \sum_{p \nmid m} \chi(a^{-1}) \chi(p) p^{-s} = \sum_{p \nmid m} \sum_{\chi} \chi(a^{-1}p) p^{-s} = \sum_{p \nmid m} p^{-s} \sum_{\chi} \chi(a^{-1}p)$$

where we swap the sums by finiteness. By orthogonality, we have the inner sum equal to 0 unless $a^{-1}p \equiv 1 \pmod{m}$ in which case the sum is $\varphi(m)$.

Hence

$$g_a(s) = \varphi(m) \sum_{p \equiv a \pmod{m}} p^{-s}.$$

Taking $s \rightarrow 1^+$, by Lemma 52 we have terms in the $g_a(s)$ sum bounded for all χ except $\chi = 1$. Thus $g_a(s) \sim f_1(s) \sim -\log(1-s) \sim \sum_p p^{-s}$, and thus the analytic density of $\mathbb{P}_a$ is $\frac{1}{\varphi(m)}$. $\square$

Proposition 55. If $f(n)$ is the proportion of primes up to the n th prime that are $a \pmod{m}$ then

$$\liminf f(n) \leq 1/\varphi(m) \leq \limsup f(n).$$

Observation 56. This does not show that the natural density of primes congruent to $a \pmod{m}$ is $\frac{1}{\varphi(m)}$, which is, by definition, equalities in Proposition 55.

Corollary 57. There are an infinite number of primes congruent to $a \pmod{m}$.

7 Appendix

Proposition 58. For odd primes p , and $a, b, c \in \mathbb{Z}/p\mathbb{Z}$ with $a, b \in (\mathbb{Z}/p\mathbb{Z})^\times$ there is some x, y such that $c = ax^2 + by^2$.

Proof. The sets ax^2 and $c - by^2$ both have $\frac{p+1}{2}$ elements. Where they intersect we have $ax^2 = c - by^2 \implies ax^2 + by^2 = c$. $\square$

Proposition 59. If p is odd and $a, b \in \mathbb{Z}$ then $(a, b)_p = 1$ iff $ax^2 + by^2 \equiv 1$ has a non-trivial solution in $\mathbb{Z}/p\mathbb{Z}$.

Proof. Observe that not both a and b can be multiples of p , so we will assume WLOG that $p \nmid a$. If $x \equiv 0 \pmod{p}$ then b is a square mod p , and by the cyclicity of $\mathbb{Z}/(p^n)\mathbb{Z}$ it is a square mod p^n . Hence, we will assume that ax is invertible mod p . We will Hensel lift our given solution.

Suppose $ax^2 + by^2 \equiv 1 \pmod{p^n}$. Then $ax^2 + by^2 \equiv 1 + cp^n \pmod{p^{n+1}}$. Let $d \in \mathbb{Z}$ such that $2xda \equiv -c \pmod{p}$. Then

$$a(x+dp^n)^2 + by^2 \equiv ax^2 + ad^2p^{2n} + 2axdp^n + by^2 \equiv ax^2 + by^2 + (2axd)p^n \equiv 1 + cp^n - cp^n \equiv 1 \pmod{p^{n+1}}.$$

This lifts our solution, and thus we are done. $\square$

Corollary 60. Let p be an odd prime. We have $(p, q)_p = \left(\frac{p}{q}\right)$ and if $\gcd(a, p) = \gcd(b, p) = 1$ then $(a, b)_p = 1$.

Theorem 61 (Hilbert). Recall that V is the set of all positive rational primes and infinity. For all $a, b \in \mathbb{Q}^\times$ we have

$$\prod_{v \in V} (a, b)_v = 1.$$

Proof. Since the symbols are bilinear on $\mathbb{Q}^\times$, it suffices to show the result for when a, b are primes or -1 .

(1), $a = b = -1$:

I claim that $(-1, -1)_\infty = (-1, -1)_2 = -1$ and $(-1, -1)_p = 1$ for odd primes p . When $p = 2$ we need a sum of 3 squares to be equal to 0, but our classification of squares (Theorem 28) disallows this.

(2), $a = -1, b = 2$:

We have $1^2 = (-1)(1)^2 + (2)(1)^2$, so $(-1, 2)_v = 1$ for all v .

(3), $a = b = p > 2$:

When $v \neq 2, p$ we have $(a, b)_v = 1$ by Corollary 60. We will show that

$$(a, b)_2 = (-1)^{\frac{p-1}{2}} = (a, b)_p.$$

We will first prove the second equality. When $p \equiv 1 \pmod{4}$ we have $p = a^2 + b^2$ and thus $p^2 = pa^2 + pb^2$, so $(p, p)_p = 1$. When $p \equiv 3 \pmod{4}$ we know that there is no solution mod p and thus no possible solution in $\mathbb{Q}_p$.

For the first equality, suppose we have a solution. Then $px^2 + py^2 = z^2$. If $z = 0$ then x/y squares to -1 . But $x^2 + 1$ has no roots mod 4, and thus has no roots in $\mathbb{Q}_2$. Hence, if we divide by z^2 we are looking a solution to $x^2 + y^2 = p$. This has no solutions mod 4 when $p \equiv 3 \pmod{4}$, and clearly has a solution when $p \equiv 1 \pmod{4}$.

(4), $a = -1, b = p > 2$:

By Proposition 22, we have $(b, b) = (-1, b)$, so this reduces to case (2) or case (3).

(5), $a = 2 < p = b$:

When $v \neq 2, p$ we have $(a, b)_v = 1$ by Corollary 60. We will show that

$$(2, p)_2 = (-1)^{\frac{p^2-1}{8}} = \left(\frac{2}{p}\right) = (2, p)_p.$$

The second equality is well-known, and we have the third equality by Corollary 60.

For the first equality, suppose that $2x^2 + py^2 = z^2$. The left side is a sum of something of odd 2-valuation and even 2-valuation, so $z \neq 0$. Hence we have a solution to $2x^2 + py^2 = 1$. If we take both sides mod 8 then we have $py^2 = 1$ or $py^2 = -1$, and since $y^2 \equiv 1 \pmod{8}$ we have $p \equiv \pm 1 \pmod{8}$. Now suppose $p \equiv \pm 1 \pmod{8}$. Then we have a solution to $2x^2 + py^2 = 1 \pmod{8}$.

We will now lift this solution. Suppose $2x^2 + py^2 \equiv 1 \pmod{2^n}$. Now we have $2x^2 + py^2 \equiv 1, 1 + 2^n \pmod{2^{n+1}}$. If it is 1 then we are done. Otherwise we have $2x^2 + p(y + 2^{n-1})^2 \equiv 2x^2 + py^2 + 2p2^{n-1} + 2^{2n-2} \equiv 1 + 2^n + p2^n \pmod{2^{n+1}}$. We have p odd, and thus $1 + 2^n + p2^n \equiv 1 \pmod{2^{n+1}}$. Hence, we can lift our solution.

(6), $2 < a = p < q = b$:

By Corollary 60, when $v \neq 2, p, q$ we have $(a, b)_v = 1$ and

$$(p, q)_p = \left(\frac{q}{p}\right) \text{ and } (p, q)_q = \left(\frac{p}{q}\right).$$

Hence, by quadratic reciprocity, it suffices to show that

$$(p, q)_2 = (-1)^{\frac{p-1}{2} \frac{q-1}{2}}.$$

Suppose $px^2 + qy^2 = z^2$. If $z = 0$ then we have $-p/q$ a perfect square, which occurs iff p/q is 7 mod 8, thus 3 mod 4, and thus (exactly) one of p and q is 3 (resp. 1) mod 4.

Hence, we assume that z is non-zero. If we divide through by z we get a solution to $px^2 + qy^2 = 1$. If we take both sides mod 8 then we see that x and y cannot both be odd, for then the left side is even, and thus one of them is even. If we assume WLOG that x is even then $px^2 \equiv 0, 4 \pmod{8}$. So thus $qy^2 \equiv 1 \pmod{4}$, and since y is odd we have $q \equiv 1 \pmod{4}$.

For the backwards direction, suppose that either q or p is 1 mod 4. WLOG suppose that $p \equiv 1 \pmod{4}$. We will show that $px^2 + qy^2 = 1$ has a solution mod 2^n , and that we can lift it. Observe that it has a solution mod 8 because either $p \equiv 1 \pmod{8}$ or $p \equiv 5 \pmod{8}$ and we take $(x, y) = (1, 0)$ and $(1, 2)$ respectively.

Now suppose that $px^2 + qy^2 \equiv 1 \pmod{2^n}$. Now we have $px^2 + qy^2 \equiv 1, 1 + 2^n \pmod{2^{n+1}}$. If it is 1 then we are done. Otherwise we have

$$p(x + 2^{n-1})^2 + qy^2 \equiv px^2 + qy^2 + 2px2^{n-1} + p2^{2n-2} \equiv 1 + 2^n + px2^n \pmod{2^{n+1}}.$$

We have px odd (observe that we have x odd by induction), and thus $1 + 2^n + px2^n \equiv 1 \pmod{2^{n+1}}$. Hence, we can lift our solution.

These 6 cases cover the $3 + \binom{3}{2}$ possibilities, and thus we have covered every case. $\square$

Note. This proof over (the field of fractions of) $\mathbb{Z}$ is ultimately just a restatement of quadratic reciprocity. Over rings of integers of number fields this same result holds and is non-trivial.

Proposition 62. A form of rank $n \geq 5$ represents 0 over $\mathbb{Q}_p$.

Proof. If $n > 5$, pick all but 5 coordinates to be 0, and we reduce to the $n = 5$ case. Observe that some g of rank 2 represents $a \in \mathbb{Q}_p^\times$, iff $(a, -d) = \varepsilon$. Hence, g takes on at least 2 values. Hence, g represents some element $a \in \mathbb{Q}_p^\times / (\mathbb{Q}_p^\times)^2$ other than d , and thus $f \sim aX^2 + h$ for h of rank 4. By definition, the discriminant of h is $d/a \neq 1$. Thus, by the $n = 4$ proof (Theorem 30) we have that h represents 0, and taking $X = 0$, we are done. $\square$